

Flight Trial Demonstration of Secure GBAS via the L-band Digital Aeronautical Communications System (LDACS)

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INTRODUCTION

The ground-based augmentation system (GBAS) is used to improve the accuracy of global navigation satellite system (GNSS) to allow GNSS-based instrument landings of aircraft. It is based on reference stations with known positions at the airport, which generate correction data and integrity parameters from GNSS measurements. Correction and integrity data are transmitted to approaching aircraft. Based on these corrections, aircraft can calculate their position with precision and confidence in the integrity of the solution. GBAS enables modern aircraft to perform safe and secure GNSS-based landings while offering several advantages over the instrument landing system (ILS) commonly used today: Approaches no longer need to be carried out in a straight line, but can also be curved and with flexible and higher glide angles; it is cheaper than ILS while, at the same time, being transparent in use to the aircrew [1].

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GBAS requires a datalink to transmit GNSS correction data to the on-board avionics of the aircraft. As of now, this datalink is specific to GBAS: The VHF data broadcast (VDB) datalink.

With GBAS evolving to GBAS approach service type (GAST)-D, which shall enable CAT III landings, VDB becomes a limitation: First, current versions of the VDB datalink broadcast only corrections for the L1 frequency of GPS satellites, which is problematic in terms of availability, especially in equatorial zones, due to ionospheric disturbances of GPS reference measurements based on a single frequency [2]. This can be alleviated by extending GBAS with a multifrequency and multiconstellation approach, including also L5 frequencies and the GALILEO, GLONASS, and BeiDou GNSS systems [3], [4]. However, the VDB datalink is a bottleneck for this extension. It does not provide sufficient throughput for correction and integrity data for multiple constellations and frequencies [5], [6]. Large and complex airports aggravate this issue: In case a single VDB transmitter cannot provide coverage of the complete airport, two transmitters using alternate time slots on the same VDB channel need to be installed, halving datalink capacity. Second, the current VDB service range of 42 km has been criticized as being too small. The criticism is that for pilots to verify that the system is operational; an increased range of GBAS services would reduce the pressure and stress level during the final approach. Third, cyber-resilience is also problematic with VDB, since it provides no cyber-security measures on par with modern wireless systems [7], [8].

These drawbacks show clearly that GBAS would benefit from a more capable datalink supporting its unimpeded evolution. A very good candidate for a capable GBAS broadcast datalink is the L-band digital aeronautical communications system (LDACS). Currently,



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under international civil aviation organization (ICAO) standardization in the project team terrestrial data link (PT-T) [9], LDACS is the future aeronautical datalink for applications related to the safety and regularity of flight for terrestrial long-range communications [10].

The objective of this article is to demonstrate an experimental in-flight implementation of GBAS using the LDACS as datalink 1) to multiply the available bandwidth, 2) to increase the service range, and 3) to introduce state-of-the-art broadcast cyber-security using timed efficient stream loss-tolerant authentication (TESLA) [11]. The flight trials discussed in this article were cofunded by the German national research project migration towards integrated COM/NAV avionics (MICONAV) and took place in March and April 2019. During MICONAV also other experiments were performed that are not discussed in this article.

METHODOLOGY

The main purpose of the experiments was to demonstrate the ability of LDACS to provide a secure datalink for GBAS. For this purpose, we implemented a terrestrial LDACS ground station and GBAS reference station [see Figure 2(a) and (b)], as well as a software-based GBAS receiver. We



Figure 1.
DLRs research aircraft Falcon 20-E5 (D-CMET) used in the experiments.

mounted the receiver on a research aircraft [see Figures 1 and 2(c)] and flew the setup in the vicinity of the ground station.

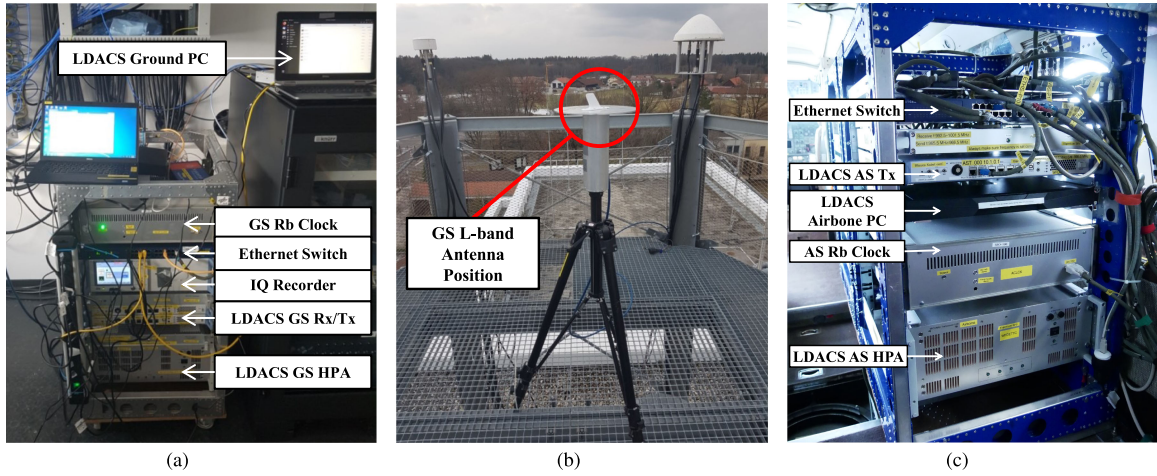
EXPERIMENTAL SETUP

The experimental setup consisted of one LDACS aircraft-station and one LDACS ground station. The GBAS ground station was colocated with the LDACS ground station. Figure 3 shows the complete experimental setup in detail: The GBAS ground station receives GNSS data with a Tallysman TW 3972 antenna, processes it using a JAVAD Sigma unit, and passes it to the GBAS ground processing software. The processed GBAS data are passed on to the ground station PC and authenticated with the TESLA protocol before it is transmitted in a message format specific to the experiment. If desired, TESLA authentication can be turned off as indicated in the red box at the center of the diagram. The ground PC is connected to the LDACS ground-station radio via user datagram protocol(UDP)/internet protocol (IP). In the aircraft-station, received messages are processed in the reverse order to verify GBAS correction data. The correction data are then used by the GBAS software in the aircraft to calculate GBAS position, velocity, and time (PVT) and protection levels.

LDACS DATALINK SETUP

The LDACS aircraft-station was configured for bidirectional communication as specified in [9] and [12]. In addition to LDACS radio equipment, the aircraft-station integrated a Ubuntu 18.04 LTS Linux computer for LDACS and GBAS airborne data processing, and several measurement and storage devices, such as a Rubidium clock, a spectrum analyzer, and an I/Q data recorder. Several of these parts are identified in Figure 2(c).

The LDACS ground station as well as the GBAS ground station and antennas were installed on top of an office building at the German Aerospace Center (DLR)

**Figure 2.**

Overview of airborne and ground LDACS equipment. (a) LDACS ground station. (b) LDACS antenna. (c) Airborne equipment.

premises near Munich, Germany. The LDACS ground station and its parts are shown in Figure 2(a) and (b). The location of the ground station is indicated in Figure 5.

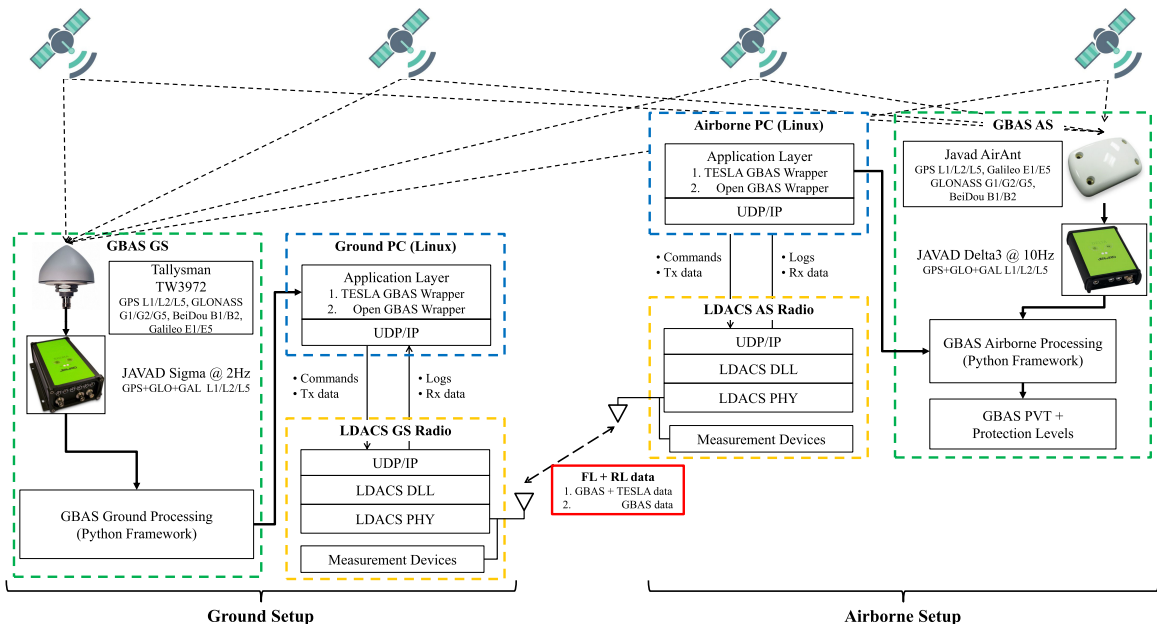
GBAS SETUP

Our experimental setup of the airborne GBAS installation on board the Falcon 20-E5 research aircraft consisted of a Javad air antenna and a Javad Delta 3 unit. This setup was used to receive and process GPS, GLONASS, and Galileo Signals on L1/L2, and L5, and provide data to GBAS airborne processing where the data were combined with the correction data received via LDACS to calculate GBAS PVT and protection levels.

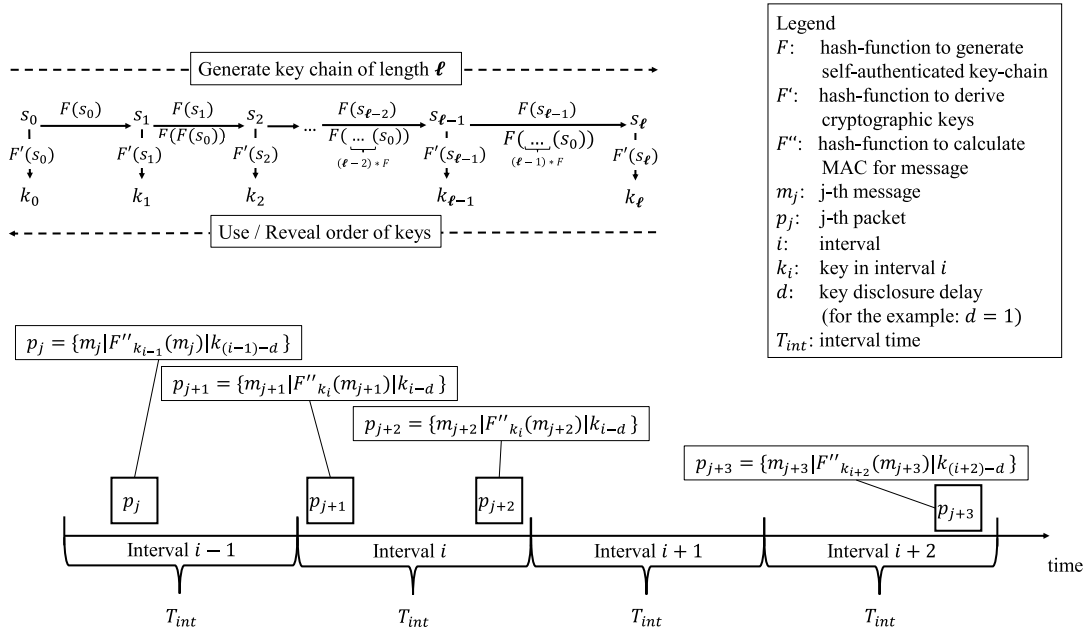
The GBAS ground station used a Tallysman TW3972 antenna. A Javad Sigma unit was used to process the GPS, GLONASS, and Galileo Signals on L1/L2, and L5 and send the data to GBAS ground processing. This setup is based on the setup discussed in [13]. GBAS air and ground processing was implemented in software in a *python3* application running on Ubuntu 18.04 LTS Linux.

DATA PROCESSING

GBAS ground processing generates correction and integrity data from the received GNSS signals. These data are forwarded to the LDACS ground-station software running

**Figure 3.**

Secure GBAS experimental setup.

**Figure 4.**

Overview of the TESLA broadcast authentication key-chain generation, key use order, and the construction and content of TESLA packets.

on the ground-station computer. GBAS data can be transmitted to the aircraft in two ways.

- 1) GBAS data can be authenticated with the TESLA protocol to secure GBAS. The secured GBAS message is then transmitted to the aircraft via LDACS.
- 2) GBAS data can be left without TESLA protection. In this case, the GBAS message is immediately transmitted to the aircraft via LDACS.

The (1) TESLA secured or (2) plain GBAS data are transmitted from the ground station to the airborne station via LDACS radio messages. The messages received by the LDACS airborne radio are processed and delivered via UDP/IP to the processing Ubuntu 18.04 LTS Linux computer on board the plane. The data are (1) verified and—if cryptographic authentication succeeded—delivered to the GBAS airborne processing framework. If the GBAS data were not authenticated, they are (2) just extracted from the received LDACS messages and immediately forwarded to GBAS airborne processing. GBAS airborne processing combines the received correction and integrity data with the GNSS signals captured by the Javad Air antenna and the Javad Delta-3 receiver to calculate GBAS PVT and protection levels.

TESLA BROADCAST AUTHENTICATION

For the experiment, we implemented a version of the TESLA broadcast authentication protocol to secure GBAS communication over the airgap between ground station and aircraft station. TESLA was first proposed by Perrig *et al.* [11], uses symmetric keys, requires time

synchronization between sender and receiver and the ability to buffer messages on the receiver.

Let us assume our communication partners to be Alice (the LDACS ground station) and Bob (the LDACS aircraft-station). For TESLA, we use three functions: a hash function F , to generate a self-authenticated key-chain, a key derivation function F' to derive cryptographic keys for message authentication code (MAC) calculations, and another hash function F'' to calculate MACs for a message m . First, Alice splits time in equal intervals T_{int} . Alice then generates a self-authenticated key-chain (e.g., by choosing a random start value and then applying a suitable hash function F iteratively) and assigns each key s_i to its respective interval i of length T_{int} . Then, via a key derivation function F' and with key s_i as input, cryptographic keys k_i for MAC calculations are generated for every interval i (e.g., $F'(s_i) = k_i$). The MACs are calculated on the messages m_j sent out in their respective interval i using function F'' and key k_i and message m_j as input (e.g., $MAC_j = F''_{k_i}(m_j)$). Both steps, the generation of a key-chain and construction of packets p_j , are depicted in Figure 4. When Alice sends out messages, Bob can immediately receive them and buffer them until he can verify the MAC. Alice delays the publication of the key k_i , which is required for the calculation of the MAC of message m_j in interval i by a certain time interval d . With that, Alice ascertains that at time T_i —the time the message was generated and sent by Alice—only she knew the key for the calculation of the MAC of that message. Releasing the key k_i d intervals later, Bob can verify the correctness of the MAC of previously received and buffered messages. Verification of correctness of MAC_j

proofs to Bob (or any other party receiving the broadcast message), that the message m_j in interval i with MAC_j was actually sent by Alice, since no one else knew the key k_i at that time T_i . The content of the different packets p_j with the respective key in the respective interval contained in the packet is again depicted in Figure 4.

For Bob to partake in the TESLA protocol, Alice and Bob need to synchronize their clocks within a margin of acceptable error [11]. Then, Alice sends TESLA parameters such as functions F , F' , and F'' , the time interval schedule consisting of interval duration T_{int} , start time T_0 , index of interval i and the length of the one-way chain, the key disclosure delay d , and a key commitment to the key chain, allowing Bob to verify that the received keys are actually part of the key-chain. These parameters need to be distributed in an authenticated manner. In the experiment, public keys and certificates of the ground station and aircraft station were bilaterally exchange via LDACS. In an operational deployment of LDACS, the distribution of public keys and certificates will likely be realized via an LDACS specific public key infrastructure, as described in [14]–[17]. Knowing Alice's public key, Bob can verify the authenticity of the TESLA parameters and start buffering messages sent by Alice until he receives the correct key to verify their authenticity.

Note that TESLA authentication requires the buffering of received messages until Alice's authentication key has been received. This introduces a key disclosure delay increasing the communication latency between Alice and Bob.

Time synchronization between aircraft station and ground station was implemented as described in [11]. The exchange of TESLA parameters was signed via an Ed25519 digital signature of the ground station. Our implementation used *python3* and the *nacl* [18] crypto-library with F being the SHA-512 hash function for key stream generation and two variations F' , F'' of the *blake2b* hash function for MAC key derivation and MAC generation.

LIMITATIONS OF THE EXPERIMENTAL SETUP

Our experiments used a single GBAS ground receiver with limited ground monitoring, thus without ionosphere and ephemeris monitoring or B -value checks. However, this has no influence on the purpose of the experiment; thus, it characterizes GBAS over LDACS similar as proposed in [8].

GBAS corrections and integrity parameters were generated and broadcast for GPS, Galileo, and GLONASS for L1 100 s and L5 100 s processing modes. As we used the LDACS message format, we did not transmit final approach segment (FAS) data and did not use the VDB message format. Transmission of all data was combined in one correction/integrity message per epoch. Producing corrections for all visible satellites our experimental setup generated data rates as predicted in [8]: Approximately 3500 B/s, distributed over several communication packets per second. To allow for better characterization of LDACS

Table 1.

Flight Trajectories and Experiments.				
Exp.	Trajectory	Duration (s)	Security	Parameters
01	Flight 1	762.160	TESLA	$T_{int} = 1s, d = 1$
02	Flight 1	3755.519	TESLA	$T_{int} = 1s, d = 2$
03	Flight 1	1459.296	TESLA	$T_{int} = 1s, d = 2$
04	Flight 2	1213.501	unsecured	improved msg. format
05	Flight 2	2765.582	unsecured	improved msg. format
06	Flight 2	3028.761	TESLA	$T_{int} = 1s, d = 2$ improved msg. format

performance, each set of corrections (every 0.5 s) was sent twice with the redundant second message 0.2 s delayed.

The position of the antenna on the aircraft was in an unfavorable place between the wings under the belly of the aircraft. This was due to the availability of port-holes in the experimental aircraft (c.f. Figure 1) and resulted in short outages discussed ahead.

FLIGHT TRAJECTORIES AND EXPERIMENTS

We performed six experiments in two flights. We chose two different flight trajectories to demonstrate secure GBAS via LDACS with different pitch and roll alignments of the aircraft-station and ground-station antennas. During the first flight, we transmitted only TESLA-secured GBAS via LDACS. We varied the key verification delay to compare different TESLA parameters for GBAS performance. In the second flight, the first two experiments were conducted with unsecured GBAS via LDACS, whereas the last experiment used TESLA again. Flight 2 used a more efficient GBAS message format. The experiments are summarized in Table I. Both flights included considerable taxiing times and preparation times on the apron not included in the table.

The first flight took place on March 26, 2019, had takeoff at 08:53 UTC, touches down at 10:50 UTC, and was chosen as dedicated test flight to demonstrate secure GBAS. Its total airtime was 7000 s with a distance of 1048 km covered. Our goal was to climb, remain at a constant altitude of 6000 m for as long as possible to have different pitch and roll configurations toward the ground-station antenna at constant altitude, and then descend and land, which we achieved as plotted in Figure 5.

The second flight trajectory was used to directly compare TESLA-secured GBAS via LDACS and unsecured GBAS via LDACS in several experiments during the same flight. We chose a greater distance to our ground station, two different flight altitudes, steeper and longer curves, and missed approaches provoking and resulting in more antenna

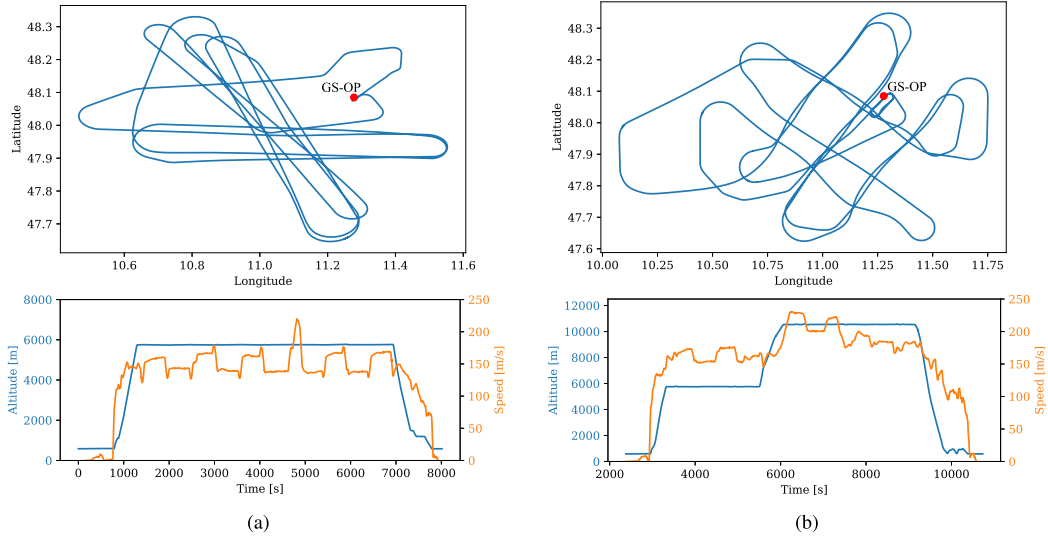


Figure 5.

Trajectory, speed, and altitude of secure GBAS via LDACS experimental flights. The location of the ground station is marked “GS-OP” for Ground-Station Oberpfaffenhofen near Munich, Germany. (a) Flight 1. (b) Flight 2.

shadowing. We achieved all of these prerequisites, as shown in Figure 5. Takeoff was on April 2, 2019 at 14:03 UTC, touch down at 16:06 UTC, with a total flight time of 7424 s and 1291 km covered with the trajectory depicted in Figure 5(b).

RESULTS

The quality of GBAS is determined by its availability. We use “GAST-C” and “GAST-D” availability according to RTCA DO-253D [19].

We start with a detailed view around the airport, measuring the GBAS correction age in Figure 6. The figure shows GBAS correction age during arrival, departure (flight 1), three go-arounds (flight 2), and taxiing. Throughout taxiing and after takeoff, still close to the airport, we experienced

stable performance with GBAS data correction ages well below the requirements (GAST-D: 1.5 s, GAST-C: 3.5 s) with a single short (10 s) LDACS outage during one of the go-arounds. Note that taxiways and runways are covered although the ground-station antenna is shadowed by a hangar building. Figure 6(b) shows the availability of the GBAS solutions over the entire course of flight 2. We see that in most situations during flight 2, we had GAST-D coverage implying a GBAS correction age of less than 2 s. Analyzing the median and 95% percentile ($P_{95\%}$) of the correction age in Table 2, these observations are confirmed.

Note that in experiments 01–03, one GBAS message was fragmented into five LDACS packets. In experiments 04–06, a more efficient message format was used. No fragmentation was needed in these experiments. This resulted in lower correction ages for flight 2.

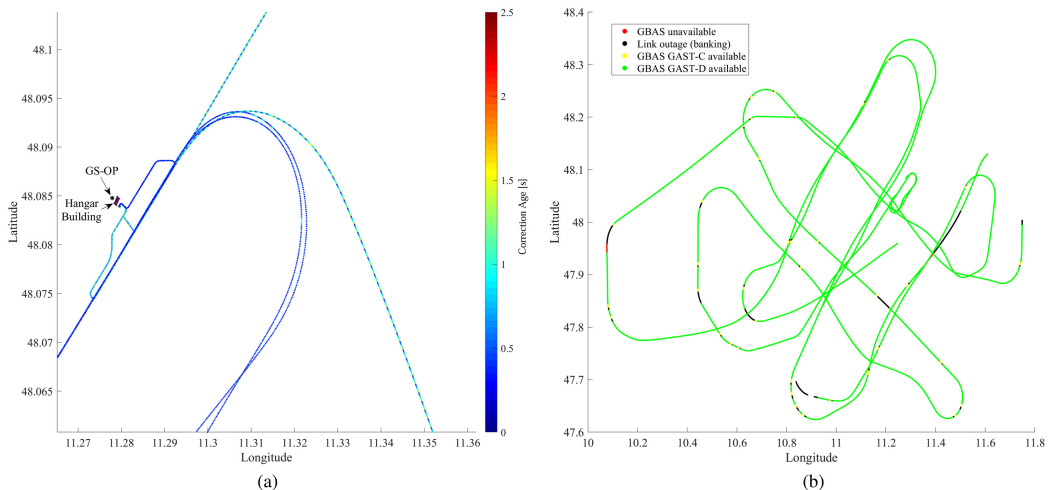


Figure 6.

(a) Age of GBAS correction data around EDMO airport. (b) GBAS availability during flight 2.

Table 2.

GBAS via LDACS Correction Age and Availability.					
Exp.	Security	Median	$P_{95\%}$	GAST-C	GAST-D
		Cor. Age	Cor. Age	Availability	Availability
Flight 1 (GBAS message fragmented in 5 LDACS packets)					
01	after TESLA ($d = 1$)	1.8 s	2.2 s	88.17%	87.40%
	before authentication	0.9 s	1.3 s	99.92%	99.84%
02	after TESLA ($d = 2$)	2.8 s	3.3 s	97.91%	97.90%
	before authentication	0.9 s	1.3 s	99.40%	99.40%
03	after TESLA ($d = 2$)	2.8 s	3.2 s	99.30%	99.30%
	before authentication	0.9 s	1.3 s	99.84%	99.84%
Flight 2 (GBAS message not fragmented)					
04	unsecured	0.4 s	0.6 s	99.98%	99.98%
05	unsecured	0.4 s	0.6 s	99.66%	99.61%
06	after TESLA ($d = 2$)	2 s	2.6 s	97.61%	97.61%
	before authentication	0.4 s	0.6 s	99.97%	99.97%

Note that experiments 04 and 05 do not employ TESLA. In all other experiments GBAS availability has been measured before and after TESLA authentication.

INCREASED DATALINK THROUGHPUT

Our experiments generated and transmitted GBAS correction and integrity data for GPS, GLONASS, and Galileo on L1/L2, and L5. The measured offered load and throughput on the LDACS data link is presented in Table 3. Our results indicate that depending on the

GBAS, update rates 31–50 kb/s were required for GBAS GAST-D services in dual-frequency and multi-constellation mode. The difference in data load between flights 1 and 2 is a result of a change of the experimental GBAS message format. Note that the generated GBAS data also depend on the number of visible satellites. Taking into account that the minimum data rate offered by LDACS is 300 kb/s in the ground-to-air direction, LDACS offers more than enough capacity to support GBAS, air traffic services (ATS), or aeronautical operational control (AOC) on the same channel.

Table III.

GBAS Over LDACS Datalink Throughput.				
Exp.	Duration in (s)	Avg. Offered Load in (kb/s)	Avg. Throughput in (kb/s)	Loss in %
Flight 1				
01	762.160	49.80	49.31	0.99
02	3755.519	45.56	44.94	1.37
03	1459.296	48.10	47.93	0.36
Flight 2				
04	1213.501	34.58	34.43	0.43
05	2765.582	34.19	33.86	0.94
06	3028.761	31.69	31.37	1.01

INCREASED SERVICE RANGE

The LDACS specification [12] stipulates at a maximum averaged equivalent isotropically radiated power (EIRP) of 52 dBm resulting in maximum designed communication range of 200 NM. However, local regulations allowed us to transmit only an EIRP of 40 dBm in our experiments. This reduction of 12 dBm—assuming free-space path loss—should result in a reduced communication range of at least 50 NM (92.6 km).

During the testing of GBAS in flights 1 and 2, we reached a maximum distance of 61 km and 94 km, respectively. On both occasions, we could send and receive GBAS packets, making the maximum demonstrated GBAS via LDACS service range in our experiment 94 km. As the service volume of GBAS contains 42 km around the local ground facility

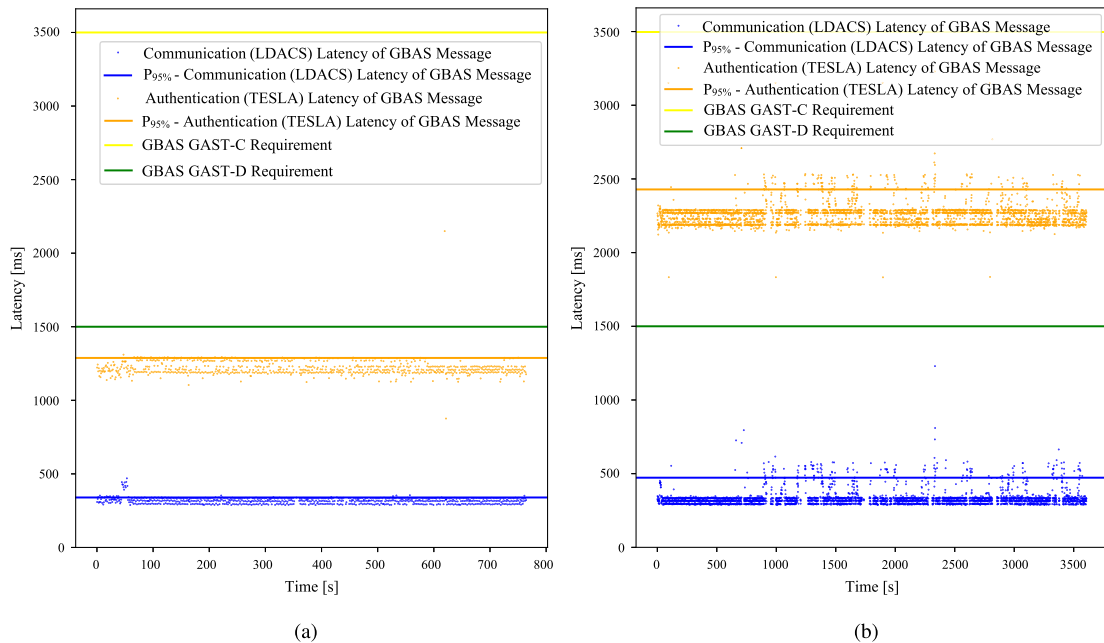


Figure 7.

Latencies of secured GBAS messages. Note that the GBAS messages in experiments 01 and 02 have been fragmented into five LDACS packets. Latency can be improved with the more efficient message format used in flight 2. (a) Experiment 01 ($T_{\text{int}} = 1\text{ s}$, $d = 1$). (b) Experiment 02 ($T_{\text{int}} = 1\text{ s}$, $d = 2$).

with the VDB datalink, we demonstrated the possibility to double that distance with 12 dBm EIRP less than foreseen for an operational LDACS deployment.

CYBER-SECURITY

TESLA security adds a configurable key disclosure delay to the authentication latency. We chose conservative key disclosure delays for our experiments and set the interval length to 1 s ($T_{\text{int}} = 1\text{ s}$) and the key delay to one time interval ($d = 1$) for experiment 01 and two intervals ($d = 2$) for all other experiments.

Changing the key delay resulted in different authentication latency measurements. Secure GBAS messages sent in experiment 01 experienced approximately 1 s less latency. This verifies the correctness of the implementation of the TESLA protocol, as in experiment 01, all messages can already be verified when they have received the key in the next message ($d = 1$), whereas in all other experiments, the right key is contained in the message after the next message ($d = 2$). The 1-s difference in authentication latency is clearly visible in Figure 7(a) and (b). Note that the latency values of these figures are not directly comparable to the correction ages presented in Table 2. The GBAS correction age is measured relative to the start of the GBAS epoch, the communication latency is not.

TESLA cyber-security adds additional load to the datalink. In our configuration, it added 144 B per secured GBAS message. This resulted in the average security data overhead of 14.4% per TESLA secured message. Time

synchronization and parameter exchange during the setup of TESLA added additional 192 B to each experiment.

DISCUSSION

Safety and security are strongly interrelated in aviation, which make strong cyber-security, a key enabler for digitalization in aviation [20]. Unfortunately, cyber-security for aeronautical communication is still not realized in most deployed systems [21], [22]. Current aeronautical digital terrestrial datalinks in use, such as VDB, have a low data rate of 28.416 kb/s (and half of that if alternating slots need to be used) and no state-of-the-art cyber-security features [20], [22], [23].

In our experiments, we demonstrated with LDACS a viable alternative to VDB offering at least ten times the net capacity in data rate and state-of-the-art cyber-security measures. With TESLA on top of LDACS, we have shown furthermore that additional dedicated broadcast message authentication is realistic and feasible. *An important result is that all TESLA authenticated messages, which reached the aircraft, could be verified.* This showed the good choice of TESLA as broadcast authentication protocol, as all GBAS messages secured via TESLA could be verified to have been sent by an authentic ground station. This and our analysis of throughput and latency show that TESLA provides good security for GBAS via LDACS with reasonable security overhead and acceptable latency. Our results also indicated that overhead can be reduced by

better time synchronization, optimized packet formats, and more lightweight hash functions.

The latency of LDACS is sufficient to enable time-shifted broadcast authentication protocols, such as TESLA. However, we have seen that the parameter choice of TESLA is important here. *Up to a delay of 1 s, the latency for TESLA-secured GBAS messages via LDACS remains low enough to stay within the requirements for GAST-C and GAST-D services.* Tweaking the time interval and key delay times might lead to even shorter latencies than demonstrated in our experiments.

LDACS offers at least a range of 94 km for GBAS service. With higher LDACS transmission power, this range could be drastically increased to 200 NM. However, depending on the region and the local properties of the ionosphere, GBAS correction data may not necessarily be useful more than 100 NM away from the reference station [24], [25]. However, the current GBAS service volume of 42 km was criticized as too small in the past. The argument is that the corrections are not used beyond the service volume point for landing service but from the intercept point to the final approach, but for the pilot to verify that the system is already operational. An increased range of GBAS services would reduce the pressure and stress level of the pilots during the final approach, as they can check availability and operational correctness of the GBAS service significantly earlier than today.

The transmission of GBAS correction data can be limited in range or signal quality due to siting problems of VDB, as VDB requires a line-of-sight connection for expected system behavior. LDACS offers some improvements here, too. On large complex airports such as Frankfurt, one VDB transmitter cannot ensure coverage on all runways due to buildings blocking the signal. Today, those airports work with two VDB transmitters on the same channel, which halves the capacity due to alternating slots. A different technical solution not halving the capacity of the data link is very much desirable. This solution must ensure coverage on all runways for the rollout guidance up to 12 ft above the ground. *With our demonstration of GBAS via LDACS on the apron, taxiway, and runway, we demonstrated that LDACS can provide a solution for transmitting GBAS correction data reliably even under nonline-of-sight conditions.*

CONCLUSION

Our experiments demonstrated that multiconstellation, multifrequency GBAS can be realized efficiently and securely via LDACS.

Also, LDACS may provide additional benefits to GBAS: It offers enough data-rate to broadcast cryptographically secured GBAS data while offering spare capacity for other ATS or AOC services on the same channel. We have shown that LDACS can extend the GBAS service range into the order of 100 km. And we have seen that GBAS via LDACS

works in nonline-of-sight scenarios with approximately the same GBAS availability as measured during flight. We demonstrated the possibility to protect broadcast data with the broadcast authentication protocol TESLA. Our results show that TESLA is well suited for securing GBAS data via LDACS; however, TESLA parameters have to be chosen carefully.

The advantages of GBAS over LDACS—(1) increased data rate, (2) increased range, and (3) cyber-security—clearly indicate that further development of secure GBAS over LDACS may be a key enabler for the future evolution of GBAS.

NOMENCLATURE

DLR	German Aerospace Center
EIRP	Equivalent Isotropically Radiated Power
FAS	Final Approach Segment
GAST	GBAS Approach Service Type
GBAS	Ground Based Augmentation System
GNSS	Global Navigation Satellite System
ICAO	International Civil Aviation Organization
ILS	Instrument Landing System
LDACS	L-band Digital Aeronautical Communications System
MAC	Message Authentication Code
MICONAV	Migration towards Integrated COM/NAV Avionics
PVT	Position, Velocity and Time
TESLA	Timed Efficient Stream Loss-tolerant Authentication
UDP	User Datagram Protocol
IP	Internet Protocol
VDB	VHF Data Broadcast

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REFERENCES

- [1] M. Felix, T. Dautermann, and H. Becker, "GBAS landing system-precision approach guidance after ILS," *Aircraft Eng. Aerosp. Technol.*, vol. 85, pp. 382–388, Aug. 2013.
- [2] J. Lee and M. Kim, "Optimized GNSS station selection to support long-term monitoring of ionospheric anomalies for aircraft landing systems," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 1, pp. 236–246, Feb. 2017.

- [3] M.-S. Circiu *et al.*, “Evaluation of GPS L5, Galileo E1 and Galileo E5a performance in flight trials for multi frequency multi constellation GBAS,” in *Proc. ION GNSS*, Sep. 2015, pp. 897–906.
- [4] D. Gerbeth, M. Felux, M.-S. Circiu, and M. Caamano, “Optimized selection of satellite subsets for a multi-constellation GBAS,” in *Proc. Int. Tech. Meeting Inst. Navigation*, Feb. 2016, pp. 360–367.
- [5] M. Stanisak, A. Lipp, and T. Feuerle, “Possible VDB formatting for multi-constellation/multi-frequency GBAS services,” in *Proc. 28th Int. Tech. Meeting Satell. Division Inst. Navigation*, Sep. 2015, pp. 1507–1518.
- [6] T. Feuerle, M. Stanisak, S. Saito, T. Yoshihara, and A. Lipp, “GBAS interoperability and multi-constellation/multi-frequency trials,” in *Proc. ENRI Int. Workshop. ATM/CNS*, Jun. 2019, pp. 162–174.
- [7] J. García, “Broadband connected aircraft security,” in *Proc. Integr. Commun., Navigation, Surveillance Conf.*, Apr. 2015, pp. 1–23.
- [8] M. Felux, T. Gräupl, N. Meurer, and M. Stanisak, “Transmitting GBAS messages via LDACS,” in *Proc. AIAA/IEEE 37th Digit. Avionics Syst. Conf.*, Sep. 2018, pp. 1–7.
- [9] International Civil Aviation Organization (ICAO), “Finalization of LDACS Draft SARPs,” Tech. Rep., Working Paper WP05 including Appendix, Second Meeting of DCIWG (Data Communication Infrastructure Working Group), Montreal, CA, Oct. 2018.
- [10] M. Schnell, U. Epple, D. Shutin, and N. Schneckenburger, “LDACS: Future aeronautical communications for air-traffic management,” *IEEE Commun. Mag.*, vol. 52, no. 5, pp. 104–110, 2014.
- [11] A. Perrig, R. Canetti, J. D. Tygar, and D. Song, “The TESLA broadcast authentication protocol,” *RSA Cryptobytes*, vol. 5, pp. 2–13, Summer/Fall 2002.
- [12] T. Gräupl, C. Rihacek, and B. Haindl, “LDACS A/G specification,” *SESAR2020 PJ14-02-01 D3.3.030*, German Aerosp. Center (DLR), Oberpfaffenhofen, Germany, Dec. 2019.
- [13] M.-S. Circiu, M. Felux, D. Gerbeth, M. Caamano, and M. Meurer, “Assessment of different dual-frequency dual-constellation GBAS processing modes based on flight trials,” in *Proc. ION GNSS*, Sep. 2016, pp. 3197–3207.
- [14] N. Mäurer and A. Bilzhause, “A cyber-security architecture for the L-band digital aeronautical communications system (LDACS),” in *Proc. AIAA/IEEE 37th Digit. Avionics Syst. Conf.*, Sep. 2018, pp. 1–10.
- [15] N. Mäurer and C. Schmitt, “Towards successful realization of the LDACS cyber-security architecture: An updated datalink security threat- and risk analysis,” in *Proc. 19th Integr. Commun., Navigation, Surveillance Conf.*, Apr. 2019, pp. 1A2/1-1A2-13.
- [16] N. Mäurer, T. Gräupl, and C. Schmitt, “Evaluation of the LDACS cybersecurity implementation,” in *Proc. AIAA/IEEE 38th Digit. Avionics Syst. Conf.*, Sep. 2019, pp. 1–10.
- [17] N. Mäurer, T. Gräupl and C. Schmitt, “Comparing different Diffie-Hellman key exchange flavors for LDACS,” in *Proc. AIAA/IEEE 39th Digit. Avionics Syst. Conf.*, Oct. 2020, pp. 1–10.
- [18] D. J. Bernstein, “Cryptography in NaCl,” *Netw. Cryptography Library*, vol. 3, 2009, Art. no. 385.
- [19] Radio Technical Commission for Aeronautics (RTCA), “Minimum operational performance standards for GPS local area augmentation system airborne equipment,” Tech. Rep. DO-253D, RTCA, Washington, DC, USA, Jul. 2017. [Online]. Available: <https://standards.globalspec.com/std/10168693/RTCA%20DO-253>
- [20] B. Mahmoud, A. Pirovano, and N. Larrieu, “Aeronautical communication transition from analog to digital data: A network security survey,” *Comput. Sci. Rev.*, vol. 11, pp. 1–29, May 2014.
- [21] A. Costin and A. Francillon, *Ghost in the Air(Traffic): On Insecurity of ADS-B Protocol and Practical Attacks on ADS-B Devices*. Las Vegas, NV, USA: Black Hat USA, Aug. 2012, pp. 1–10.
- [22] M. Niraula, J. Graefe, R. Dlouhy, M. Layton, and M. Stevenson, “ATN/IPS security approach: Two-way mutual authentication, data integrity and privacy,” in *Proc. 18th Integr. Commun., Navigation, Surveillance Conf.*, Apr. 2018, pp. 1A3-1-1A3-17.
- [23] O. Osechas, M. Mostafa, T. Gräupl, and M. Meurer, “Addressing vulnerabilities of the CNS infrastructure to targeted radio interference,” *IEEE Aerosp. Electron. Syst. Mag.*, vol. 32, no. 11, pp. 34–42, Nov. 2017.
- [24] S. Pullen and P. Enge, “An overview of GBAS integrity monitoring with a focus on ionospheric spatial anomalies,” *Indian J. Radio Space Phys.*, vol. 36, pp. 249–260, Aug. 2007.
- [25] S. Saito *et al.*, “Ionospheric delay gradient model for GBAS in the Asia-Pacific region,” *GPS Solutions*, vol. 21, pp. 1937–1947, Oct. 2017.